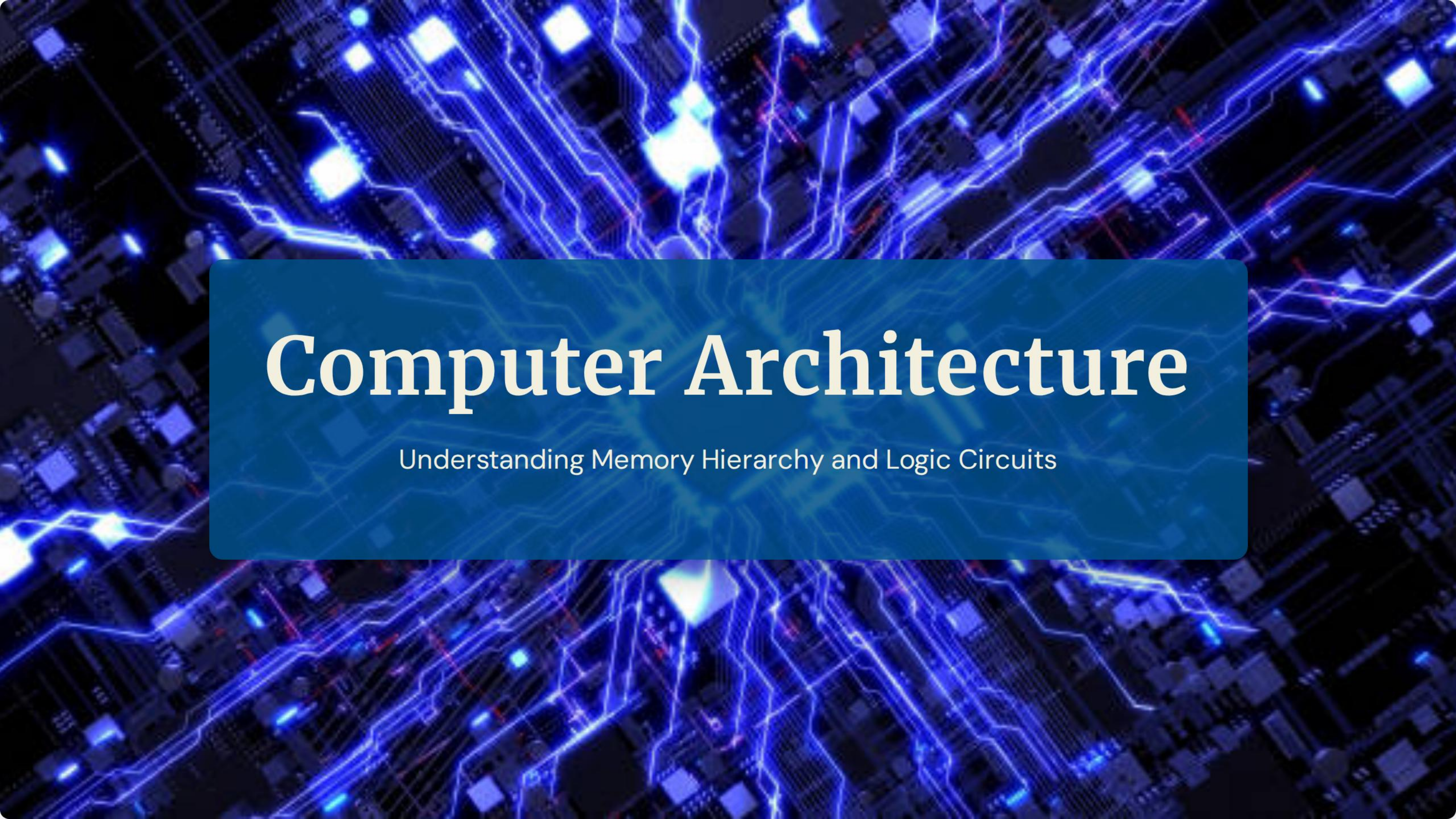




# Computer Architecture

Understanding Memory Hierarchy and Logic Circuits

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# Memory Hierarchy & Locality

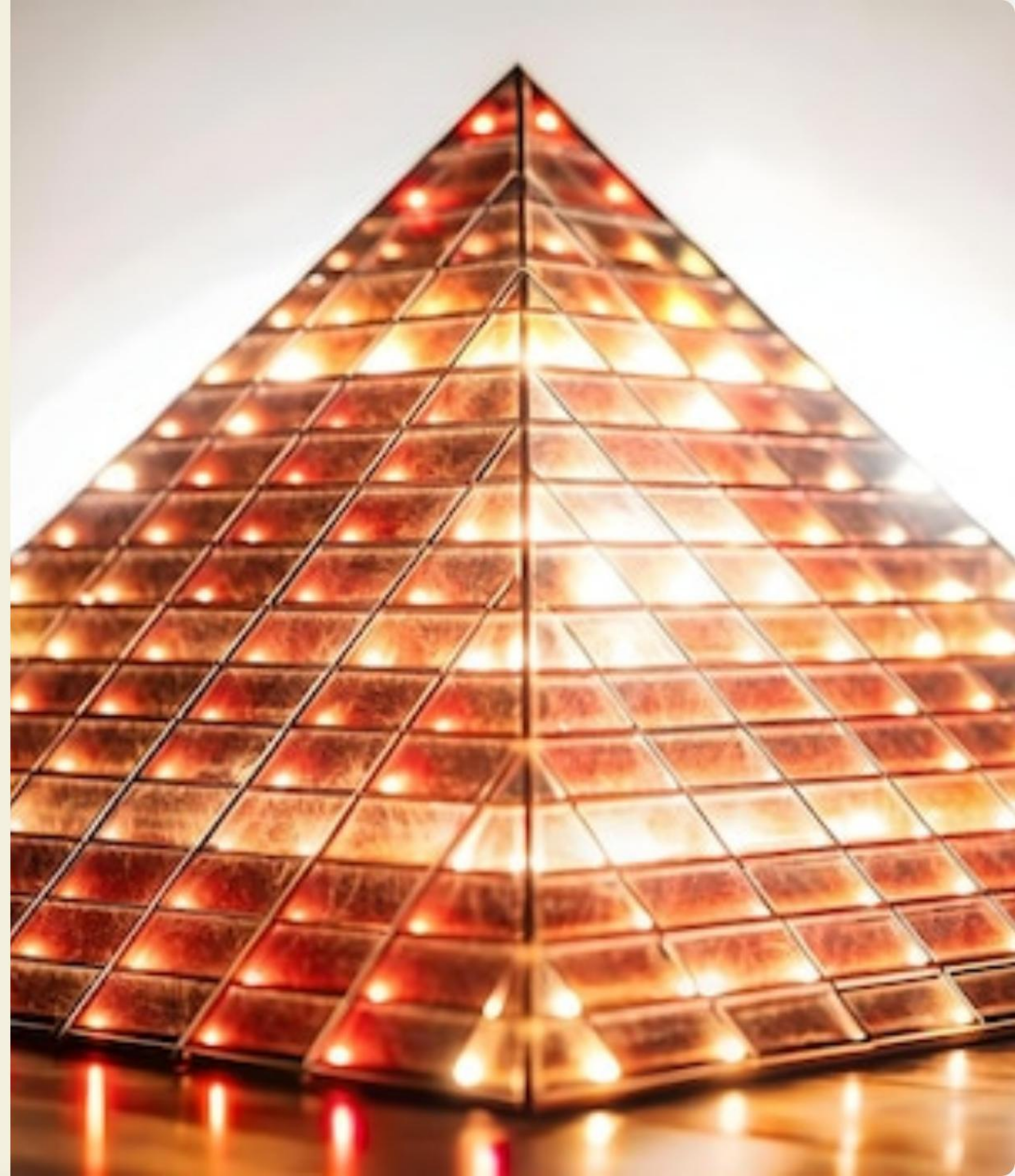
# What is Memory Hierarchy?

A computer uses different memory types to store data, arranged in a hierarchy based on speed, size, and cost.

Fast memory is quick, small, and expensive. Slow memory is larger and cheaper. This arrangement balances cost and performance optimally.

## Core Principle:

*The closer memory is to the CPU, the faster the speed but the smaller the size.*



# Levels of Memory Hierarchy



## Registers

Built directly into the CPU. The fastest and smallest memory holding active instructions and intermediate data.



## Cache (L1, L2, L3)

A high-speed buffer that reduces CPU wait time from RAM by storing frequently used data locally.



## RAM (Main Memory)

The computer's active workspace. Fast, volatile memory holding the OS, active applications, and running files.



## Virtual Memory

A section of secondary storage used as extra RAM to prevent crashes when physical RAM is overloaded.



## Secondary Storage

HDDs and SSDs. Permanent, non-volatile, slowest to access, but offers the largest long-term capacity.

# Locality of Reference

## Temporal Locality

The tendency for a program to reuse the same data or instructions within a short time frame.

*Example:* A program repeatedly updating an average temperature variable from a sensor reading every few seconds.

## Spatial Locality

The tendency for a computer to use data or instructions that are physically close to recently accessed locations.

*Example:* Reading data sequentially through an array, or reading a book page by page, allowing cache block pre-loading.

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# Logic Circuits & Minimisation

# Why Simplify Logic?

Every Boolean function expression should be reduced to its simplest form before hardware implementation. This offers several critical advantages:

- Significantly reduces the number of logic gates required.
- Increases the overall speed of circuit operations.
- Decreases power dissipation and energy consumption.
- Simplifies circuit design and lowers structural complexity.



# Representing Boolean Functions

## SOP (Sum of Products)

Combines product (AND) terms using OR operations.  
Focuses entirely on evaluating logic '1' outputs using  
Minterms.

$$f(A, B) = AB + A'B$$

## POS (Product of Sums)

Combines sum (OR) terms using AND operations.  
Focuses entirely on evaluating logic '0' outputs using  
Maxterms.

$$f(A, B) = (A + B) \cdot (A' + B)$$

# Logic Gates in Architecture

| Component                   | Role of Logic Gates                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Arithmetic Logic Unit (ALU) | Performs core arithmetic (add, subtract) and logical operations using combinations of XOR, AND, and OR gates.                 |
| Registers                   | Built from flip-flops (made of NAND/NOR gates) to temporarily store single bits and instructions during processing.           |
| Memory Units                | Forms latches/flip-flops to store bits. Gate combinations create decoders for addressing and regulate read/write permissions. |

# Core Processing Components



## ALU

The processing engine acting as the calculator for the entire computer, performing mathematical and bitwise operations.



## Registers

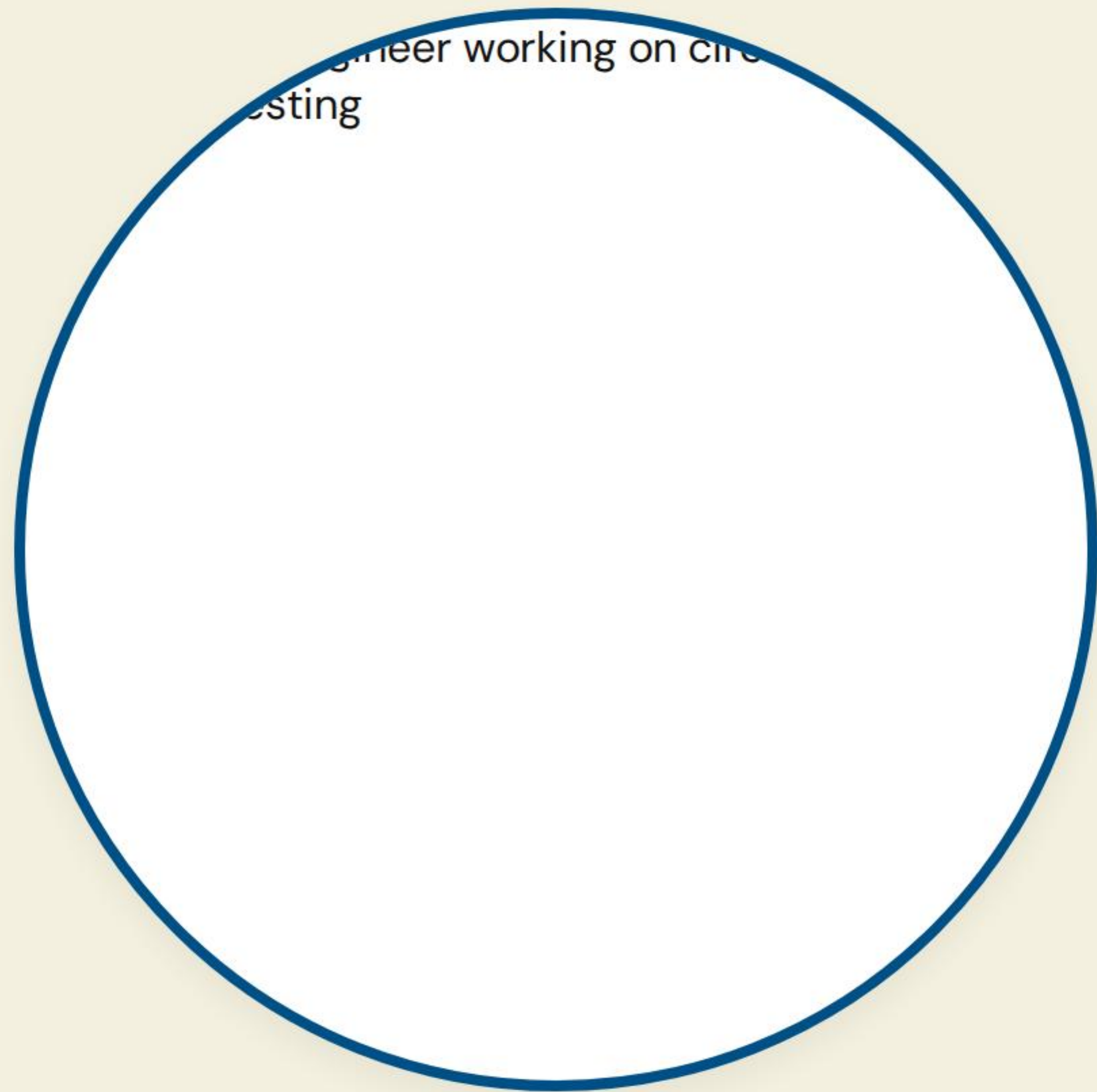
Immediate access memory embedded in the CPU, holding data currently being manipulated by the ALU.



## Memory Unit

Provides the fundamental mechanism for storing data bits, utilizing decoders to pinpoint specific cell addresses.

# Simulation vs. Hardware



## Two Approaches to Logic

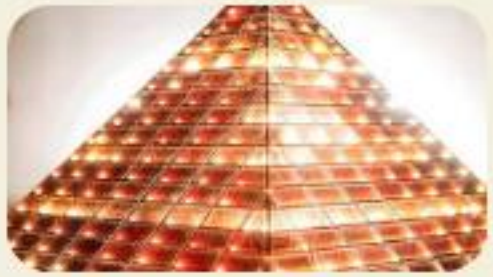
**Simulation** uses a software model (like Logisim or Verilog) to represent a logic circuit. It allows designers to test, verify, and debug safely and cost-effectively.

**Hardware implementation** means building the physical circuit using real components (ICs, FPGAs). Circuit behavior here is influenced by physical factors like voltage levels, noise, and propagation delay.

# Questions?

Thank you for exploring Computer Architecture.

# Image Sources



[https://img.freepik.com/premium-photo/flat-glowing-abstract-pyramid-representing-organizational-hierarchy-concept-as-overhead-shot\\_980716-580921.jpg](https://img.freepik.com/premium-photo/flat-glowing-abstract-pyramid-representing-organizational-hierarchy-concept-as-overhead-shot_980716-580921.jpg)

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